

US and UK. In contrast, the most recent version of Humlum and Vestergaard (2025) found minimal effects on entry-level earnings or hours worked in Denmark.¹¹

3 Data Description

3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several restrictions for our main analysis sample. We include only workers with positive earnings, exclude part-time employees, and subset to people under age 70.¹²

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms by keeping only companies that have employee earnings records for each month from January 2021 through September 2025.

ADP observes job titles for about 70% of workers in its system. We exclude workers without a recorded job title. There are over 7,000 standardized job titles, including “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of these job titles to a 2010 Standard Occupational Classification (SOC) code, additionally using information such as the job description, industry, location, and other relevant data. We use these estimated SOC codes to merge our data to occupational AI exposure measures.

After these restrictions we have records on 3.5 and 5 million workers each month for our main analysis sample, though we consider robustness to alternative analyses such as allowing for firms to enter and leave the sample.

While the ADP data include millions of workers in each month, the distribution of firms using

¹¹An important direction for future work is to reconcile differences among these recent studies, assessing the extent they arise from differences in labor market institutions, measurement of AI adoption, statistical methodology, or other factors.

¹²While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.

ADP services does not exactly match the distribution of firms across the broader US economy. Further details on differences in firm composition can be found in [Cajner et al. \(2018\)](#) and [ADP Research \(2025\)](#).¹³

3.2 Occupational AI Exposure

We use two approaches for measuring occupational exposure to AI. The first uses exposure measures from [Eloundou et al. \(2024\)](#), who estimate AI exposure by O*NET task using ChatGPT validated with human labeling. They construct occupational exposure measures by aggregating the task data to the 2018 SOC code level. We focus on the GPT-4 based β exposure measures from their paper.

The second approach we take uses generative AI usage data from the Anthropic Economic Index ([Handa et al., 2025](#)). This index reports the estimated share of queries pertaining to each O*NET task based on several million conversations with Claude, Anthropic’s generative AI model. It aggregates the data to the occupational level based on these task shares. One feature of the Anthropic Economic Index is that for each task it reports estimates of the share of queries pertaining to that task that are “automotive,” “augmentative,” or none of the above. We use this as an estimate of whether AI usage for an occupation is primarily complementary or substitutable with labor.¹⁴

We use a 2010 SOC code to 2018 SOC code crosswalk from the BLS to merge the exposure measures to the payroll data. Table [A1](#) shows example occupations for each AI exposure measure.

3.3 Other Data

To compare employment changes for teleworkable versus non-teleworkable occupations, we use data from [Dingel and Neiman \(2020\)](#). We use the Personal Consumption Expenditure index from the BLS to compute real earnings, indexed to October 2017. We use monthly Current Population Survey (CPS) data as a comparison for our main findings.

¹³[Cajner et al. \(2018\)](#) find a somewhat higher share of manufacturing and services firms compared to the QCEW using data from March 2016. They also find that ADP somewhat overrepresents firms in the Northeast. In addition, firms using ADP tend to grow faster on average than the typical firm in the US economy.

¹⁴[Handa et al. \(2025\)](#) use Claude to classify conversations into six categories. Directive and Feedback Loop conversations are considered Automotive; Task Iteration, Learning, and Validation are considered Augmentative. They also instruct the model to choose None of the above “liberally.” Table [A2](#) reproduces Table 1 from [Handa et al. \(2025\)](#) and shows more details about the automation and augmentation measures.

occupation cells. Future work should study employment trends in other large-scale data sources such as the American Community Survey (ACS) or data from Revelio Labs as in [Hosseini and Lichtinger \(2025\)](#) and [Klein Teeselink \(2025\)](#).

5 Conclusion

We document six facts about the recent labor market effects of artificial intelligence.

- We find substantial employment declines for early-career workers in occupations most exposed to AI, such as software development and customer support.
- While economy-wide employment continues to grow, employment growth for young workers has been stagnant.
- Entry-level employment has declined in applications of AI that *automate* work, with muted effects for those that *augment* it.
- After conditioning on firm-time effects, young workers experienced a 16% relative employment decline in the most exposed occupations.
- Labor market adjustments are more visible in employment than in compensation.
- These patterns hold across various alternative analyses.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment significantly.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in adjustment periods as workers reallocate from displaced forms of work to new forms with growing labor demand ([Autor et al., 2024](#)). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others ([Bresnahan et al., 2002](#); [Brynjolfsson et al., 2021](#)).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. We will continue monitoring these outcomes to assess whether the trends

documented in the paper accelerate in the future. Future work would benefit from better firm-level AI adoption data, which would provide sharper variation for estimating plausible causal effects of AI on employment.